

GAS-MPC: pre-fixed-task-set results

Reference only — superseded by the fixed 200-task-set results in `main.tex`

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Why this document exists

Every result below used a per-seed random pair set: seed s drew its own (start, goal) pairs via `default_rng(s)`, so seed-to-seed spread confounded task difficulty with CEM’s own sampling noise. This is why the phase-6 protocol switched to one fixed 200-task set shared by every seed (see `main.tex`, §§ “Fixed task set”). Everything here predates that switch and is kept only as a reference for how the earlier sweeps (objective choice, graph/lookahead knobs, executor fixes, critic weight and combinations) narrowed down to the finalists that were then re-run under the fixed-task protocol. Do not cite success rates from this document as the headline numbers — use `main.tex` instead.

Per-seed pair sets (50 pairs per seed)

All tables in this block: each seed draws its own 50 (start, goal) pairs; within a seed every configuration sees the same pairs. Cells are success rates in %, mean $\pm$ sd over the seeds tested (per-seed counts and values are in the appendix). Default settings unless a row says otherwise: lookahead 13.7 TDR units, $H_{TD} = 8$, TE 0.9, open-loop 25 steps, no retrieval.

Table 1: L2.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	88.0 $\pm$ 0.0	38.7 $\pm$ 8.2	18.0 $\pm$ 4.0	9.6 $\pm$ 1.5
retrieval	–	50.0 $\pm$ 0.0	–	10.0 $\pm$ 0.0

Table 2: L2 + critic.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
$\beta=1$	–	50.0 $\pm$ 0.0	–	12.0 $\pm$ 0.0
$\beta=2$	–	–	–	18.0 $\pm$ 0.0

Table 3: TDR (no graph).

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	84.0 $\pm$ 0.0	48.0 $\pm$ 0.0	28.0 $\pm$ 0.0	10.0 $\pm$ 0.0

Table 4: CTG (terminal).

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	84.0 ± 0.0	50.0 ± 0.0	28.0 ± 0.0	10.0 ± 0.0
retrieval	–	–	–	14.0 ± 0.0

Table 5: subgoal.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	88.0 ± 0.0	56.0 ± 0.0	–	16.0 ± 0.0
retrieval	–	–	–	20.0 ± 0.0

Table 6: subgoal + critic.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
$\beta=1$	–	–	–	26.0 ± 0.0

Table 7: subgoal_tdr.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
lookahead=8	–	50.0 ± 0.0	–	–
replan=1	–	–	–	14.0 ± 0.0
replan=2	–	36.0 ± 0.0	–	18.0 ± 0.0
replan=2, retrieval	–	–	–	6.0 ± 0.0
default	84.0 ± 0.0	55.3 ± 2.5	30.0 ± 0.0	10.0 ± 1.6
retrieval	–	54.0 ± 0.0	–	24.0 ± 0.0
lookahead=20	–	56.0 ± 0.0	–	–
$H_{TD}=12$	–	64.0 ± 0.0	–	8.0 ± 0.0
$H_{TD}=16$	–	50.0 ± 0.0	–	–

Table 8: subgoal_tdr + critic.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
$\beta=0.5$	–	–	–	14.0 ± 0.0
$\beta=1$	–	72.0 ± 0.0	58.0 ± 0.0	28.0 ± 5.5
retrieval, $\beta=1$	–	64.0 ± 0.0	–	32.4 ± 12.9
$H_{TD}=12$, $\beta=1$	–	56.0 ± 0.0	–	22.0 ± 0.0
$TE=0.999$, $\beta=2$	–	–	–	24.0 ± 0.0
$TE=0.99$, $\beta=2$	–	–	–	22.0 ± 0.0
$\beta=2$	–	66.7 ± 7.5	46.0 ± 6.0	33.6 ± 5.1
retrieval, $\beta=2$	–	–	–	30.0 ± 0.0
$\beta=4$	–	64.0 ± 0.0	50.0 ± 0.0	27.2 ± 9.0
retrieval, $\beta=4$	–	–	–	30.0 ± 2.8
$\beta=8$	–	–	–	25.6 ± 5.6

Table 9: Direction.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	78.0 ± 0.0	26.0 ± 0.0	–	8.0 ± 0.0

Table 10: Waypoint path.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	86.0 ± 0.0	42.0 ± 0.0	–	6.0 ± 0.0

Per-seed pair sets (100 pairs per seed)

All tables in this block: each seed draws its own 100 (start, goal) pairs; within a seed every configuration sees the same pairs. Cells are success rates in %, mean $\pm$ sd over the seeds tested (per-seed counts and values are in the appendix). Default settings unless a row says otherwise: lookahead 13.7 TDR units, $H_{TD} = 8$, TE 0.9, open-loop 25 steps, no retrieval.

Table 11: L2.

variant	cross-ep
default	10.0 ± 0.0

Table 12: subgoal_tdr + critic.

variant	cross-ep
retrieval, $\beta=1$	36.0 ± 0.0
$\beta=4$	36.0 ± 0.0

Per-seed results

Every test in the blocks above, broken down by seed.

Table 13: Per-seed pair sets (50 pairs), same-ep 25, per seed, and pooled over the seeds each row has.

configuration	s0	pooled
l2	88	88.0
tdr	84	84.0
ctg_htd8_te0.9	84	84.0
subgoal_la13.7_th8_htd8_te0.9	88	88.0
subgoal_tdr_la13.7_th8_htd8_te0.9	84	84.0
dir_la13.7_th8_htd8_te0.9	78	78.0
path_sugap5_th8_htd8_te0.9	86	86.0

Table 14: Per-seed pair sets (50 pairs), same-ep 50, per seed, and pooled over the seeds each row has.

configuration	s0	s1	s2	pooled
l2	40	28	48	38.7
l2_ret	50	–	–	50.0
l2_crit1	50	–	–	50.0
tdr	48	–	–	48.0
ctg_htd8_te0.9	50	–	–	50.0
subgoal_la13.7_th8_htd8_te0.9	56	–	–	56.0
subgoal_tdr_la8_th8_htd8_te0.9	50	–	–	50.0
subgoal_tdr_la13.7_th8_htd8_te0.9_rh2	36	–	–	36.0
subgoal_tdr_la13.7_th8_htd8_te0.9	58	52	56	55.3
subgoal_tdr_la13.7_th8_htd8_te0.9_ret	54	–	–	54.0
subgoal_tdr_la20_th8_htd8_te0.9	56	–	–	56.0
subgoal_tdr_la13.7_th12_htd12_te0.9	64	–	–	64.0
subgoal_tdr_la13.7_th16_htd16_te0.9	50	–	–	50.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	72	–	–	72.0
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	64	–	–	64.0
subgoal_tdr_la13.7_th12_htd12_te0.9_crit1	56	–	–	56.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	72	72	56	66.7
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	64	–	–	64.0
dir_la13.7_th8_htd8_te0.9	26	–	–	26.0
path_sugap5_th8_htd8_te0.9	42	–	–	42.0

Table 15: Per-seed pair sets (50 pairs), same-ep 100, per seed, and pooled over the seeds each row has.

configuration	s0	s1	pooled
l2	22	14	18.0
tdr	28	–	28.0
ctg_htd8_te0.9	28	–	28.0
subgoal_tdr_la13.7_th8_htd8_te0.9	30	–	30.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	58	–	58.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	52	40	46.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	50	–	50.0

Table 16: Per-seed pair sets (50 pairs), cross-ep, per seed, and pooled over the seeds each row has.

configuration	s0	s1	s2	s3	s4	pooled
l2	8	12	10	8	10	9.6
l2_ret	10	–	–	–	–	10.0
l2_crit1	12	–	–	–	–	12.0
l2_crit2	18	–	–	–	–	18.0
tdr	10	–	–	–	–	10.0
ctg_htd8_te0.9	10	–	–	–	–	10.0
ctg_htd8_te0.9_ret	14	–	–	–	–	14.0
subgoal_la13.7_th8_htd8_te0.9	16	–	–	–	–	16.0
subgoal_la13.7_th8_htd8_te0.9_ret	20	–	–	–	–	20.0
subgoal_la13.7_th8_htd8_te0.9_crit1	26	–	–	–	–	26.0
subgoal_tdr_la13.7_th8_htd8_te0.9_rh1	14	–	–	–	–	14.0
subgoal_tdr_la13.7_th8_htd8_te0.9_rh2	18	–	–	–	–	18.0
subgoal_tdr_la13.7_th8_htd8_te0.9_rh2_ret	6	–	–	–	–	6.0
subgoal_tdr_la13.7_th8_htd8_te0.9	10	12	8	–	–	10.0
subgoal_tdr_la13.7_th8_htd8_te0.9_ret	24	–	–	–	–	24.0
subgoal_tdr_la13.7_th12_htd12_te0.9	8	–	–	–	–	8.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	14	–	–	–	–	14.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	26	36	20	26	32	28.0
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	44	30	22	50	16	32.4
subgoal_tdr_la13.7_th12_htd12_te0.9_crit1	22	–	–	–	–	22.0
subgoal_tdr_la13.7_th8_htd8_te0.999_crit2	–	–	24	–	–	24.0
subgoal_tdr_la13.7_th8_htd8_te0.99_crit2	–	–	22	–	–	22.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	38	40	26	34	30	33.6
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit2	30	–	–	–	–	30.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	44	28	18	22	24	27.2
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit4	34	28	28	–	–	30.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit8	32	24	18	32	22	25.6
dir_la13.7_th8_htd8_te0.9	8	–	–	–	–	8.0
path_sugap5_th8_htd8_te0.9	6	–	–	–	–	6.0

Table 17: Per-seed pair sets (100 pairs), cross-ep, per seed, and pooled over the seeds each row has.

configuration	s0	pooled
l2	10	10.0
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	36	36.0
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	36	36.0

Per-pair diagnostic (Phase 1, cross-episode, seed 0)

Referenced from `main.tex`'s Phase 1 findings-log entry. D_g : graph geodesic from start to goal (Alg.-1 start selection, goal attached as in planning), in env steps via the TDR step-gap calibration.

Table 18: Per-pair diagnostic, cross protocol, seed 0, 50 pairs. D_g : graph geodesic from start to goal (Alg.-1 start selection, goal attached as in planning), in env steps via the TDR step-gap calibration. 'block near': initial block position error < 60 px and angle error < 30°. Cells: successes / pairs in the bin.

objective # pairs	$D_g \in [0, 50)$ 14	$D_g \in [50, 100)$ 33	$D_g \in [100, 200)$ 3	$D_g \in [200, \infty)$ 0	block near 8
ctg_htd8_te0.9	3	2	0	0	1
ctg_htd8_te0.9_ret	3	4	0	0	3
dir_la13.7_th8_htd8_te0.9	2	2	0	0	2
l2	2	2	0	0	1
l2_crit1	3	3	0	0	2
l2_crit2	3	6	0	0	1
l2_ret	3	2	0	0	1
path_sugap5_th8_htd8_te0.9	1	2	0	0	0
subgoal_la13.7_th8_htd8_te0.9	4	4	0	0	2
subgoal_la13.7_th8_htd8_te0.9_crit1	4	9	0	0	2
subgoal_la13.7_th8_htd8_te0.9_ret	4	4	2	0	3
subgoal_tdr_la13.7_th12_htd12_te0.9	2	2	0	0	1
subgoal_tdr_la13.7_th12_htd12_te0.9_crit1	3	7	1	0	5
subgoal_tdr_la13.7_th8_htd8_te0.9	3	2	0	0	2
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	4	3	0	0	4
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	3	9	1	0	3
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	7	12	0	0	6
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	6	14	2	0	5
subgoal_tdr_la13.7_th8_htd8_te0.9_ret	6	6	0	0	3
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	8	12	2	0	7
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit2	6	8	1	0	7
subgoal_tdr_la13.7_th8_htd8_te0.9_rh1	2	4	1	0	0
subgoal_tdr_la13.7_th8_htd8_te0.9_rh2	4	5	0	0	2
subgoal_tdr_la13.7_th8_htd8_te0.9_rh2_ret	2	1	0	0	2
tdr	3	2	0	0	1